Year 12 Further Maths - Mathematical induction

Similar questions - Matrix powers and inequalities

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 **In this question you must show detailed reasoning.** The matrix $\mathbf{M} = \begin{pmatrix} 1 & 4 \\ 0 & 3 \end{pmatrix}$. Prove that $\mathbf{M}^n = \begin{pmatrix} 1 & 2(3^n - 1) \\ 0 & 3^n \end{pmatrix}$ for any positive integer n. **[6]**

2 You are given that $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix}$ where a is a constant. Prove by induction that $\mathbf{A}^n = \begin{pmatrix} 1 & 0 \\ an & 1 \end{pmatrix}$ for all integers $n \geqslant 1$. **[5]**

3 Prove by induction that $2^n > n^2$ for all integers $n \geqslant 5$. **[5]**

Answers

Question 1

Base: $n=1$: $\begin{pmatrix}1 & 2(3-1)\\ 0 & 3\end{pmatrix}=\begin{pmatrix}1 & 4\\ 0 & 3\end{pmatrix}=\mathbf{M}$. *Assume* $\mathbf{M}^k=\begin{pmatrix}1 & 2(3^k-1)\\ 0 & 3^k\end{pmatrix}$. *Step:* $\mathbf{M}^{k+1}=\mathbf{M}^k\mathbf{M}=\begin{pmatrix}1 & 4+3\times 2(3^k-1)\\ 0 & 3^{k+1}\end{pmatrix}=\begin{pmatrix}1 & 2\times 3^{k+1}-2\\ 0 & 3^{k+1}\end{pmatrix}=\begin{pmatrix}1 & 2(3^{k+1}-1)\\ 0 & 3^{k+1}\end{pmatrix}$. This is the required form for $n=k+1$, so by induction true for all $n\geqslant 1$

Question 2

Base: $\mathbf{A}^1=\mathbf{A}$ ($an=a$). *Assume* $\mathbf{A}^k=\begin{pmatrix}1 & 0\\ ak & 1\end{pmatrix}$. *Step:* $\mathbf{A}^{k+1}=\mathbf{A}^k\mathbf{A}=\begin{pmatrix}1 & 0\\ ak+a & 1\end{pmatrix}=\begin{pmatrix}1 & 0\\ a(k+1) & 1\end{pmatrix}$. True for $n=k+1$; hence true for all $n\geqslant 1$

Question 3

Base: $n=5$: $32>25$. *Assume* $2^k>k^2$ for some $k\geqslant 5$. *Step:* $2^{k+1}=2\cdot 2^k>2k^2$. Now $2k^2-(k+1)^2=k^2-2k-1=(k-1)^2-2>0$ for $k\geqslant 5$. So $2^{k+1}>(k+1)^2$. Hence true for all $n\geqslant 5$ by induction